Arisha Raza

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EDUCATION

University of Georgia, School of Computing — Athens, GA
Bachelor of Science, Computer Science — August 2027
Emphasis, Software Engineering and Cybersecurity

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C#, SQL, Bash
Backend & Systems: FastAPI, .NET 8, REST APIs, SQLAlchemy, Distributed Systems
Infrastructure: Docker, AWS ECS Fargate, Azure, Terraform, GitHub Actions
Analytics & Observability: Power BI, Monte Carlo Simulation, Prometheus, SLA Monitoring
Security: Active Directory, SIEM, Network Segmentation, System Hardening
Frontend: React, Next.js, Vite, State Management

RELEVANT EXPERIENCE

Precision Medicine Group — Remote
IT Operations Intern — June 2026 - Present

- Built interactive Power BI dashboards translating enterprise service desk data into operational trends and actionable insights.
- Architected and deployed a C#/.NET 8 incident orchestration system integrating 3 Teams interfaces, Azure, and PagerDuty; automated escalations with correlation IDs and structured telemetry.
- Built a 2-stage GitHub Actions pipeline with Azure OIDC, private runners, and deployment health checks.
- Engineered Python automation converting approved ManageEngine requests into verified Copilot license assignments through Microsoft Graph.
- Built 7 security-gated workflows and 114 automated tests covering identity resolution, capacity checks, duplicate prevention, rollback, and recovery; enforced least-privilege authentication, idempotency, and post-write verification.

Style My Ride — Alpharetta, GA
Software Product Engineering — January 2024 - Present

- Engineered a React configuration engine using state management, conditional rendering, and asset layering to support 50+ boot design permutations.
- Implemented a layered rendering pipeline mapping materials to composable boot components; optimized state updates and delivered subsecond previews across 100+ combinations.
- Deployed a customer-facing configurator for $1,000–$3,000 custom boots, supporting real-time design workflows at retail and competition events.

Radar STEM Camp — Remote
Founder, Director, and Technical Program Lead — June 2020 - Present

- Founded and scaled a free virtual STEM program serving 500+ students across 15+ sessions; taught Python, web development, and introductory algorithms.
- Led 10+ volunteers developing curriculum and the program website, and delivering live sessions for cohorts of 25–40 students.
- Expanded the program to students in 4+ countries; featured in Forsyth County News for increasing access to computer science and STEM education.

PROJECTS

Smart Jump — Voice-Controlled Equestrian Digital Twin
Python, BLE Emulation, State Machines, Voice Control, GitHub Actions
GitHub: github.com/arisharazakhan-cpu/smart-jump-prototype

- Architected a voice-controlled digital twin coordinating 2 emulated BLE controllers and 3 rails across 61 height settings (12–72 inches), modeling synchronized jump adjustments.
- Engineered dual-layer fault tolerance enforcing 5 safety invariants and coordinated stops for desynchronization, heartbeat loss, telemetry failure, and unsafe commands.
- Deployed an interactive simulator with live telemetry, fault injection, event logging, and confidence-gated voice commands; validated behavior through 18 automated tests across 4 critical failure scenarios.

PitWall — Interactive F1 Race Strategy Simulator
Next.js, TypeScript, React, OpenF1 API, Monte Carlo Simulation, Vercel
GitHub: github.com/arisharazakhan-cpu/pitwall-race-strategy-simulator

- Engineered a Monte Carlo strategy engine evaluating 100,000 race outcomes in an 83.6 ms local Node.js median benchmark; modeled tyre degradation, pit timing, weather, safety cars, and traffic.
- Integrated 7 OpenF1 telemetry datasets into an interactive race-operations dashboard with pit commands, pace visualization, strategy comparisons, and temporal holdout backtesting.
- Deployed to Vercel with API validation, timeouts, cache fallback, and circuit breaking; passed 11 simulation-invariant tests and automated type-checking, production builds, and CodeQL scanning through GitHub Actions.

Decision Support System — Enterprise SLA Monitoring Service
Python, FastAPI, MySQL, SQLAlchemy, Alembic
GitHub: github.com/arisharazakhan-cpu/decision-support-system

- Built a configurable SLA evaluation service processing operational metrics across rolling windows and database-defined thresholds; distinguished passing, failing, and missing-data outcomes.
- Engineered a decision engine mapping SLA results to healthy, degraded, and critical states with HTTP 200/206/503 signaling; aggregated 4 dashboard indicators and timestamped diagnostics.
- Designed a 4-layer backend separating API routes, evaluation services, ORM models, and MySQL persistence; implemented Alembic migrations and database-aware health checks.

Cloud Native Backend Platform
Python, FastAPI, AWS ECS Fargate, Docker, Terraform, SQLAlchemy, Prometheus
GitHub: github.com/arisharazakhan-cpu/cloud-native-backend-platform

- Architected an AWS backend on ECS Fargate with Terraform-managed VPC networking, load balancing, IAM, and CloudWatch logging; containerized a FastAPI service with JWT authentication and SQLAlchemy persistence.
- Engineered CI/CD across build, publish, and deploy stages using GitHub Actions, Docker, ECR, and Terraform; pinned container images to commit SHAs for traceable releases.
- Instrumented request counters, latency histograms, and HTTP health checks; built a reproducible Docker Compose workflow demonstrating registration, authentication, resource creation, and retrieval.